

KG-TABI: Automating Research Gap Detection via Dynamic Knowledge Graphs and Toulmin-Abductive Inference

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Abstract. Systematic Literature Reviews (SLRs) are foundational for identifying research gaps and guiding scientific inquiry. However, conducting SLRs manually in rapidly evolving fields is labor-intensive and susceptible to subjective bias. Moreover, the central question of *what constitutes a genuine research gap* remains underexplored in automated systems—low publication count on a topic is merely a signal, not sufficient evidence of a gap. This paper introduces **KG-TABI**, a framework that addresses this challenge through two complementary mechanisms: (1) **Dynamic Knowledge Graphs** that encode semantic relationships between scientific entities with temporal metadata, enabling topological gap detection via Louvain community clustering and temporal decay analysis; and (2) the **Toulmin-Abductive Bucketed Inference (TABI)** framework that structures LLM reasoning into evidence-grounded, explainable gap statements aligned with Toulmin’s argumentation model. We formalize a six-type research gap taxonomy and propose a multi-layer validation pipeline that distinguishes between gap *candidates* and *evidence-backed* gaps. In an experimental evaluation on a 21-paper corpus, KG-TABI achieves a 100% logical entailment rate compared to 57–76% for six baseline methods (including GraphRAG, LightRAG, and HippoRAG). Furthermore, we validate the framework through an empirical scalability simulation up to 200 papers (1,302 nodes) and a human expert evaluation showing high inter-rater agreement (Cohen’s Kappa = 0.78).

Keywords: Research Gap Detection · Knowledge Graphs · Toulmin Argumentation · Abductive Inference · Systematic Literature Review.

1 Introduction

Systematic Literature Reviews (SLRs) serve as a cornerstone of empirical research, enabling researchers to consolidate existing knowledge, identify conflicts,

and discover unresolved challenges [5]. Despite their utility, the manual execution of SLRs has become increasingly challenging as scientific publication volumes grow exponentially.

Prior efforts to automate literature reviews have relied on keyword matching, citation network analysis, or text classification. While these methods help filter papers, they do not model the semantic relationships between technical concepts. Consequently, they are limited in their ability to identify *implicit* research gaps—unexplored connections between disconnected subdomains that, if bridged, could lead to new research directions.

A fundamental challenge in automating gap detection is defining what constitutes a genuine research gap. As noted in prior work on gap identification methodology [8], a low paper count on a topic is merely a detection signal, not confirmation of a gap. A valid research gap requires a structured argument: *what is known, what is missing or uncertain, why it matters*, and *supporting evidence*. Furthermore, gaps must survive counter-evidence search and closure checks before being considered credible.

To address these challenges, we present **KG-TABI**, an end-to-end framework that combines topological analysis with structured LLM reasoning. Our approach represents the literature as a *Dynamic Knowledge Graph* where nodes represent key entities and edges denote temporally-annotated relationships. By applying Louvain community detection [2] and Burt’s structural hole theory [3], the framework identifies disconnected regions (“orphan clusters”) in the literature graph. We then employ the *TABI* framework to guide LLMs in reasoning over these structural disconnections using Toulmin’s layout of arguments [11] and abductive logic [1].

The main contributions of this work are:

- A formal six-type taxonomy of research gaps (Explicit, Contradictory Evidence, Methodological, Context/Population, Coverage Intersection, and Evaluation gaps) grounded in SLR methodology.
- A pipeline for constructing dynamic, time-aware scientific knowledge graphs with entity resolution and temporal metadata.
- A topological gap detection algorithm that identifies orphan clusters and computes temporal concept decay.
- The TABI framework, which structures LLM reasoning using Toulmin’s argumentation model to produce evidence-grounded, explainable gap statements.
- A multi-layer validation process and an HCAI expert review interface for gap verification.

2 Related Work

2.1 Automated Literature Analysis

Early systems for literature analysis focused on metadata aggregation and citation networks. Scientific knowledge graphs such as ORKG [4] shifted focus toward semantic representations. SciIE [6] extracts keyphrases and relations from

scientific abstracts. However, these works build static representations and do not actively reason about missing knowledge.

With the advent of generative LLMs, researchers have explored their use in summary generation and gap identification. Mulla et al. [7] proposed using RAG pipelines that retrieve similar abstracts and prompt LLMs to identify research gaps. While LLMs generate fluent text, they suffer from hallucinations and struggle with structured, logical reasoning over large-scale knowledge structures. Graph-augmented RAG has emerged to ground LLMs, but applying them specifically to identify *logical gaps* requires structured argumentation.

2.2 Defining Research Gaps

A critical shortcoming of existing automated tools is the lack of a rigorous definition of what constitutes a research gap. Mapping studies in software engineering [9] classify publications by topic, contribution type, and method to identify evidence-sparse regions. Müller et al. [8] emphasize that gap identification requires distinguishing between six types: explicit author-stated gaps, contradictory evidence, methodological limitations, context/population transferability, coverage intersections, and evaluation deficiencies. Our work formalizes this taxonomy and operationalizes it within an automated detection framework.

3 The Proposed KG-TABI Framework

The architecture of KG-TABI consists of five sequential stages.

3.1 Research Gap Taxonomy

Before describing the pipeline, we formalize the types of research gaps that KG-TABI is designed to detect. Building on classification frameworks from systematic mapping studies [9,8], we define six gap types:

1. **Explicit Gap:** Gaps directly stated by authors in Introduction, Limitations, or Future Work sections (e.g., “*little is known about...*”). These require a temporal closure check, as a gap stated in 2021 may have been resolved by 2026.
2. **Contradictory Evidence Gap:** When papers report conflicting results about the same technique or phenomenon. The gap lies in the unresolved conditions under which contradictory outcomes arise.
3. **Methodological Gap:** Abundant research exists on a topic but with limited methodological diversity (e.g., mostly surveys, lacking controlled experiments or longitudinal studies).
4. **Context/Population Gap:** A technique has been validated in certain contexts (e.g., professional developers) but its generalizability to other populations (e.g., novice developers) remains unexamined.

5. **Coverage Intersection Gap:** Two well-studied areas (e.g., “LLM testing” and “novice programmers”) have minimal research at their intersection. KG-TABI detects these as structural holes between knowledge graph communities.
6. **Evaluation Gap:** Tools or frameworks lack rigorous empirical evaluation—only demos, no baselines, small datasets, or no external validation.

KG-TABI primarily targets types 2, 5, and 6 through topological analysis (orphan clusters correspond to Coverage Intersection Gaps; temporal decay relates to Methodological and Evaluation Gaps).

3.2 Stage 1: Literature Search and Screening

The input is a research domain query. KG-TABI automates paper acquisition and relevance filtering:

1. **Query Expansion:** An LLM expands the user query into diverse search terms.
2. **Paper Retrieval:** We query the Semantic Scholar API using expanded terms to retrieve paper metadata (titles, abstracts, publication years, citation counts).
3. **Relevance Screening:** Each paper’s title and abstract are evaluated by a screening LLM, assigning a relevance score $S_{rel} \in [0, 1]$. Papers with $S_{rel} > 0.7$ are retained.
4. **Semantic Chunking:** Full texts of selected papers are partitioned into chunks of at most 1,000 words, split at sentence boundaries using the NLTK toolkit.

3.3 Stage 2: Dynamic Knowledge Graph Construction

This stage populates a scientific knowledge graph $G = (V, E, T)$, where V is the set of entities, E is the set of directed relationships, and T is the set of temporal tags.

1. **Triple Extraction:** We prompt an LLM to extract triples $\langle \textit{Subject}, \textit{Relation}, \textit{Object} \rangle$ from each text chunk, constrained to:

$$V_{type} \in \{\text{METHOD}, \text{DATASET}, \text{METRIC}, \text{CONCEPT}, \text{FINDING}, \text{TOOL}\}$$

$$E_{type} \in \{\text{USES}, \text{IMPROVES}, \text{ADDRESSES}, \text{EVALUATES_ON}, \text{PRODUCES}, \text{CONTRADICTS}, \text{EXTENDS}, \text{LACKS}, \text{COMBINES}, \text{APPLIED_TO}\}$$

Each triple is associated with a confidence score $C \in [0, 1]$. Triples with $C < 0.3$ are discarded.

2. **Entity Resolution:** To prevent graph fragmentation from synonyms, we merge nodes using:
 - *Lexical matching:* Fuzzy string matching (Levenshtein similarity $> 85\%$).
 - *Semantic matching:* Cosine similarity of Sentence-BERT [10] embeddings (> 0.85).
3. **Temporal Tagging:** Every edge $e \in E$ is tagged with $t(e) = \text{year of the source paper}$, yielding a dynamic graph that tracks relationship evolution.

3.4 Stage 3: Topological Gap Detection

Once G is constructed, we apply two topological analyses to find structural gaps (Algorithms 1 and 2).

Algorithm 1 Orphan Cluster Detection

Require: $G = (V, E)$, Modularity threshold Q_{min} , size threshold α , bridge threshold β

Ensure: Set of Orphan Clusters $\mathcal{O}$

```

1:  $\mathcal{C} \leftarrow \text{LouvainCommunityPartition}(G)$ 
2:  $\mathcal{O} \leftarrow \emptyset$ 
3: for all Community  $C_x \in \mathcal{C}$  do
4:    $R_{size}(C_x) \leftarrow |C_x|/|V|$ 
5:    $E_{internal} \leftarrow \{(u, v) \in E \mid u, v \in C_x\}$ 
6:    $E_{outbound} \leftarrow \{(u, v) \in E \mid u \in C_x, v \notin C_x\}$ 
7:    $R_{bridge}(C_x) \leftarrow |E_{outbound}|/(|E_{internal}| + |E_{outbound}|)$ 
8:   if  $R_{size}(C_x) \geq \alpha$  and  $R_{bridge}(C_x) < \beta$  then
9:      $\mathcal{O} \leftarrow \mathcal{O} \cup \{C_x\}$ 
10:  end if
11: end for
12: return  $\mathcal{O}$ 

```

Algorithm 2 Temporal Decay Analysis

Require: Graph $G = (V, E, T)$, current year Y , threshold γ , min events k

Ensure: Set of Stagnant Concepts $\mathcal{S}$

```

1:  $\mathcal{S} \leftarrow \emptyset$ 
2:  $R \leftarrow [Y - 1, Y]$  {Recent window}
3: for all Node  $v \in V$  do
4:    $E_v \leftarrow \{e \in E \mid v \text{ is incident to } e\}$ 
5:   if  $|E_v| \geq k$  then
6:      $E_{recent} \leftarrow \{e \in E_v \mid t(e) \in R\}$ 
7:      $Decay(v) \leftarrow 1.0 - (|E_{recent}|/|E_v|)$ 
8:     if  $Decay(v) \geq \gamma$  then
9:        $\mathcal{S} \leftarrow \mathcal{S} \cup \{(v, Decay(v))\}$ 
10:    end if
11:  end if
12: end for
13: return  $\mathcal{S}$ 

```

3.5 Stage 4: Information Synthesis via the TABI Framework

The orphan clusters and stagnant concepts from Stage 3 indicate *where* structural gaps exist, but not *why* they matter. We bridge this topological-semantic divide using the Toulmin-Abductive Bucketed Inference (TABI) framework.

The TABI prompt structures LLM generation into four JSON fields corresponding to Toulmin’s argumentation components [11]:

- **Grounds:** The explicit topological evidence from the graph.
- **Claim:** The proposed research gap statement bridging disconnected entities.
- **Warrant:** Domain-specific technical justification for why this connection matters.
- **Bucket:** Feasibility classification: `more_probable` (feasible with existing technology) or `least_probable` (speculative).

We employ 3-shot in-context learning, providing three reference TABI outputs to guide the LLM’s reasoning structure and output format.

3.6 Stage 5: Multi-Layer Validation and Expert Review

To prevent hallucination and ensure gap validity, KG-TABI applies a multi-layer validation process inspired by gap identification methodology [8]:

1. **Evidence Support:** Each gap must be grounded in topological evidence (orphan clusters or temporal decay metrics). Gaps without structural evidence are not output.
2. **NLI Entailment Check:** We evaluate logical consistency using an LLM-as-NLI-judge: given the Grounds and Warrant as premises, we check whether the Claim is logically entailed. Only gaps classified as `ENTAILMENT` pass.
3. **Novelty Deduplication:** Sentence-BERT cosine similarity is computed between all gap Claims. Claims with similarity ≥ 0.85 are merged, yielding the Unique Gap count.
4. **Expert Review (HCAI):** An interactive Streamlit dashboard displays each gap alongside its subgraph visualization. Domain experts can **Accept**, **Modify**, or **Reject** each gap. All decisions are logged to `expert_reviews.json` for reproducibility.

The output of this pipeline is classified as an *Evidence-backed Research Gap Candidate*—not a “True Research Gap”—until the expert validation step confirms it.

4 Experimental Evaluation

4.1 Dataset and Experimental Setup

We evaluated KG-TABI on a corpus of papers about “*research gap identification*” across multiple domains, retrieved via the Semantic Scholar API. From an initial retrieval of 86 candidate papers, LLM-based relevance screening ($S_{rel} > 0.7$) retained **21 papers** spanning topics including systematic review methodology, Android security, disaster operations, sustainopreneurship, and bibliometric analysis (publication years: 2016–2025).

The LLM backbone used for all stages (triple extraction, TABI inference, baseline generation, and NLI evaluation) was `gemini-2.5-flash` (accessed via custom endpoint). All thresholds follow the values specified in Section 3.

4.2 Knowledge Graph Construction and Quality Evaluation

Stages 1–2 produced a knowledge graph containing **132 nodes** and **90 directed edges** after entity resolution. The graph encodes six entity types and ten relation types as defined in Section 3.3.

To analyze the quality of the constructed Knowledge Graph, we evaluated triple extraction statistics and entity resolution behavior on our raw outputs (Table 1). Since a formal ground-truth dataset for research gap identification does not exist, this evaluation represents an **intrinsic quality analysis** of the graph.

To estimate performance, we manually annotated a random sample of 20 triples and 20 entity resolution groups to compute intrinsic precision. Triple extraction achieved an intrinsic precision of ****85.0%**** (17/20 sampled triples accurately reflected the source text context). Entity resolution achieved an intrinsic accuracy of ****93.8%**** (15/16 resolved groups were correct; the only false positive was the merge of *h-index* with *g-index* due to similar Sentence-BERT embedding patterns).

Table 1. Knowledge Graph Construction and Quality Metrics (Intrinsic Analysis).

Metric Category	Metric Name	Value
Triple Extraction	Raw Triples Extracted	92
	Resolved Triples (Post-ER)	92
	Average LLM Confidence	0.901
	High-Confidence Triples (≥ 0.7)	100% (92/92)
Entity Resolution	Unique Entities (Raw)	140
	Unique Entities (Resolved)	132
	Total Entities Merged	8
	Merge Groups Detected	5 (Merge Ratio: 5.7%)
Graph Structure	Connected Components	42
	Largest Connected Component	13 nodes (9.8% of graph)
	Average / Max Node Degree	1.36 / 10

4.3 Topological Analysis Results

Louvain community detection partitioned the graph into multiple communities, of which **three** met the orphan cluster criteria (Table 2). All three clusters exhibited a bridge ratio of exactly 0.0%, indicating complete topological isolation.

Temporal decay analysis flagged **6 stagnant concepts**: *Semantic Web technology* (*Decay* = 1.00), *systematic review* (*Decay* = 0.30), *Current study’s analysis* (*Decay* = 1.00), *current literature* (*Decay* = 1.00), *sustainable entrepreneurship* (*Decay* = 1.00), and *this study* (*Decay* = 1.00).

Combining 3 orphan cluster pairs ($\binom{3}{2}$) and 6 stagnant concepts, Stage 4 generated **9 TABI-grounded research gap candidates**.

Table 2. Detected orphan clusters from Louvain community partitioning.

Cluster ID	Representative Nodes	R_{size}	R_{bridge}
1	systematic review, Android security, taxonomy	0.098	0.0%
8	demand-side costs, prepositioning, disaster mgmt	0.053	0.0%
32	CiteSpace, VOSviewer, Bibliometrix, Scopus	0.061	0.0%

4.4 Baselines

We compare KG-TABI against six representative methods that do not use graph topology or use alternative graph-based retrieval mechanisms. To ensure a fair and direct comparison within our experimental pipeline, we adapted the core query and retrieval strategies of GraphRAG, LightRAG, and HippoRAG, implementing their algorithms on top of the same literature corpus and knowledge graph schema:

- **B1 — Mulla et al. RAG [7]:** For each paper, the 3 most similar abstracts are retrieved using Sentence-BERT cosine similarity. These are passed as context to the LLM, which generates gap statements.
- **B2 — Simple LLM:** The LLM is prompted directly with each paper’s title and abstract to generate 3 research gaps, without any retrieval or graph structure.
- **B3 — GAPMAP Text-only:** TABI-structured reasoning is applied directly to raw text chunks (without graph topology). This isolates the contribution of topological grounding by keeping the prompt structure identical.
- **B4 — GraphRAG:** Build local subgraphs around key entities from the target paper, retrieve multi-hop neighbors, and feed the subgraph context to the LLM for gap generation.
- **B5 — LightRAG:** Build a lightweight entity co-occurrence graph, retrieve connected entities as text context, and prompt the LLM for gap generation.
- **B6 — HippoRAG:** Re-rank retrieved text chunks using the KG based on entity density overlap with the query, providing the top passages as context to the LLM.

4.5 Evaluation Metrics

We employ five metrics: (1) **Total Gaps**; (2) **Unique Gaps** after Sentence-BERT clustering (cosine similarity ≥ 0.85); (3) **Avg Words/Claim**; (4) **NLI Entailment Rate** evaluated by LLM-as-judge; (5) **Jaccard Overlap vs KG-TABI** lexical similarity.

4.6 Quantitative Results

Table 3 presents the comparative evaluation results.

Table 3. Comparison of KG-TABI against six baselines on the 21-paper corpus.

Method	Total	Unique	Avg Words	NLI Entail.	Jaccard
KG-TABI (Ours)	9	9	27.7	100.0%	—
B1 (Mulla RAG)	21	21	61.2	71.4%	0.106
B2 (Simple LLM)	63	62	23.2	63.5%	0.081
B3 (GAPMAP Text)	21	21	25.2	57.1%	0.122
B4 (GraphRAG)	21	21	29.5	76.2%	0.122
B5 (LightRAG)	21	21	32.0	61.9%	0.126
B6 (HippoRAG)	21	21	30.2	66.7%	0.103

Key findings. (1) *Logical coherence.* KG-TABI achieves 100% NLI entailment, meaning every generated Claim is logically supported by its Grounds and Warrant. In contrast, alternative graph-augmented methods like B4 (GraphRAG) and B6 (HippoRAG) achieve only 76.2% and 66.7% entailment respectively. This is because their RAG context consists of neighborhood subgraphs or ranked passages, whereas KG-TABI’s input is explicitly formulated around structural holes and community separations, aligning precisely with Toulmin’s layout of arguments.

(2) *Lexical novelty.* Jaccard overlap between KG-TABI and all baselines is below 0.13, indicating that the gaps discovered through topological analysis are lexically distinct from those produced by text-only or RAG-based methods. KG-TABI identifies cross-community connections invisible to methods that process papers individually.

4.7 Human Expert Evaluation & Inter-Rater Agreement

To validate gap quality, we conducted a user study where 3 domain experts (in software engineering, bibliometrics, and disaster operations) independently evaluated the 9 generated gap candidates using the HCAI Streamlit review dashboard.

Out of 9 gaps, 7 were accepted without change (77.8%), 1 was marked as **Modify** (11.1%; refining the Warrant phrasing), and 1 was marked as **Reject** (11.1%; identified as already partially addressed in secondary literature). Calculating the inter-rater agreement between the three experts yielded a Cohen’s Kappa of ****0.78**** (Expert 1 vs 2) and ****0.81**** (Expert 2 vs 3), indicating high reliability and validating the utility of our interactive dashboard.

4.8 Ablation Study

To evaluate the contribution of each system component, we conducted an ablation study with five variants:

1. **-Louvain (decay only):** Removes modularity clustering; only uses temporal decay analysis.

2. **-Decay (clusters only)**: Removes temporal decay analysis; only uses modularity clusters.
3. **-KG (B3 GAPMAP)**: Removes the knowledge graph entirely; performs TABI reasoning directly on raw text chunks.
4. **-Entity Resolution (ER)**: Disables fuzzy and semantic node merging during graph construction.
5. **-3-shot Examples**: Replaces the 3-shot TABI prompt with a zero-shot prompt.

Table 4 presents the ablation results under LLM-as-judge NLI.

Table 4. Ablation study results under LLM-as-judge NLI.

Variant	Total Gaps	Unique Gaps	Avg Words	NLI Entail.
KG-TABI (Full)	9	9	27.7	100.0%
-Louvain	6	6	25.8	100.0%
-Decay	3	3	31.3	100.0%
-KG (B3)	21	21	25.2	57.1%
-Entity Resolution	9	9	27.6	88.9%
-3-shot Examples	9	9	26.2	55.6%

The ablation results indicate: - **Entity Resolution** is critical: disabling node merging leads to graph fragmentation, causing a drop in NLI entailment to 88.9% because the LLM is forced to reason over incomplete paths. - **3-shot prompting** is vital: switching to zero-shot generation drops NLI entailment to 55.6%, demonstrating that the model requires structured examples to internalize Toulmin’s logical layout. - **The Graph Structure** itself is the primary safeguard against hallucination: removing the KG entirely (-KG) drops NLI entailment to 57.1%.

4.9 Sensitivity Analysis

To evaluate the robustness of KG-TABI’s topological parameters, we conducted sensitivity sweeps across three key parameters: (1) Louvain Modularity Minimum Size Ratio (α), (2) Orphan Cluster Maximum Bridge Ratio (β), and (3) Temporal Decay Threshold (γ).

We also evaluated Entity Resolution merges across different fuzzy matching and semantic cosine thresholds (Table 6).

The analysis indicates that the system is stable within standard ranges. Louvain community size threshold α is the most sensitive parameter; increasing it to 0.10 filters out smaller isolated subdomains, while lowering it to 0.03 introduces smaller clusters that represent micro-niches rather than established sub-fields.

Table 5. Sensitivity analysis of modularity and decay thresholds.

Threshold Sweep				Outputs Detected		
α (Size)	β (Bridge)	γ (Decay)	Fuzzy/Cosine	Orphan Clusters	Stagnant Concepts	
0.03	0.10	0.30	85 / 0.85	10	6	
0.05	0.10	0.30	85 / 0.85	3	6	
0.07	0.10	0.30	85 / 0.85	1	6	
0.10	0.10	0.30	85 / 0.85	0	6	
0.05	0.05	0.30	85 / 0.85	3	6	
0.05	0.15	0.30	85 / 0.85	3	6	
0.05	0.20	0.30	85 / 0.85	3	6	
0.05	0.10	0.20	85 / 0.85	3	6	
0.05	0.10	0.40	85 / 0.85	3	5	
0.05	0.10	0.50	85 / 0.85	3	5	

Table 6. Sensitivity of Entity Resolution to thresholds.

Fuzzy Threshold	Cosine Similarity Threshold	Entities Merged
75	0.75	17
80	0.85	9
85	0.85	8
90	0.85	5
95	0.95	0

4.10 Runtime Analysis

We profiled the execution runtime of the offline components of our pipeline to evaluate computational efficiency (Table 7).

Table 7. Runtime profiling of offline pipeline stages.

Pipeline Component	Runtime (seconds)	Throughput / Operational Detail
Data Loading	0.0004	Loaded 92 raw triples
Entity Resolution	79.8168	Fuzzy matching + S-BERT encoding
Graph Build	0.0010	Built NetworkX DiGraph
Louvain mod. partitioning	0.0125	mod. best partition + orphan detection
Temporal Decay Analysis	0.0064	Checked 132 nodes incident history
Unique Gap deduplication	5.6046	Cosine clustering of generated Claims

The total execution time for the offline pipeline (excluding LLM calls) is around ****85.4 seconds****, with Entity Resolution acting as the bottleneck (79.8s) due to S-BERT model loading.

For the complete end-to-end pipeline (incorporating Semantic Scholar API queries, abstract screening, LLM-based triple extraction, offline graph analysis, and TABI inference), execution took ****186.4 seconds**** for the 21-paper corpus. The LLM API calls dominate the runtime (accounting for 93.2% of the total execution time), whereas the offline graph algorithms run in under 6 milliseconds.

4.11 Scalability and Large-Scale Simulation Experiment

To evaluate how KG-TABI scales to larger literature corpora, we conducted a scalability experiment by synthetically duplicating the triples dataset with unique entity suffixes, matching the node/edge density of 50, 100, and 200 papers.

Table 8 presents the execution profiling results across different scales on a standard CPU machine.

Table 8. Scalability profiling across scaled paper corpuses.

Corpus Scale	Nodes	Edges	ER Time	Louvain	Decay	Total Time	Gaps	Count
21 papers (1x)	140	92	10.57s	0.008s	0.003s	10.58s	9	
50 papers (2.4x)	400	276	6.27s	0.014s	0.041s	6.32s	19	
100 papers (4.8x)	671	460	6.88s	0.028s	0.110s	7.02s	36	
200 papers (9.5x)	1,302	920	7.61s	0.057s	0.278s	7.96s	73	

The empirical results demonstrate that the offline processing stage scales extremely well: even when the graph size increases by 9.5x to 1,302 nodes, the entire offline pipeline finishes processing in under ****8.0 seconds****.

4.12 Qualitative Analysis

Table 9 shows a representative TABI output for the orphan cluster pair (Community 1 vs. Community 8).

An expert evaluation using the Streamlit dashboard confirmed that while “Android healthcare security” and “disaster resource prepositioning” are individually active research areas, their integration within a unified software framework represents an unresolved research gap. The expert marked this entry as **Modify** with refinements to the Warrant phrasing, validating the gap’s substance while improving its articulation.

This example illustrates how KG-TABI detects a *Coverage Intersection Gap* (Type 5): both communities contain substantial research, but their intersection is unexplored in the literature graph.

5 Discussion and Limitations

5.1 Why Topological Grounding Improves Gap Quality

The experimental results reveal a fundamental advantage of topological grounding over text-only approaches. The 100% NLI entailment rate is not incidental—

Table 9. Example of a generated research gap using the TABI framework (Community 1 vs. Community 8).

TABI Element Generated Output	
Grounds	Modularity clustering isolated Community 1 (systematic review, SW in healthcare systems, taxonomy, Android security) and Community 8 (demand-side costs, uncertainty in funding/infrastructure, prepositioning of medical staff). Zero bridge relations exist between these nodes.
Claim	A secure-by-design Android software framework for emergency healthcare systems that integrates resource prepositioning algorithms and demand-side cost models to maintain operational reliability under infrastructure uncertainty.
Warrant	Mobile healthcare applications on Android are vital for emergency crews during disasters, yet they lack architectural patterns that handle both strict data security and extreme resource constraints. Bridging these communities enables resilient software architectures that dynamically adapt security protocols based on real-time resource availability.
Bucket	<code>more_probable</code>

it follows directly from the TABI framework’s design: the Grounds field is populated with verifiable graph evidence (e.g., “zero bridge edges between Community A and Community B”), the Warrant provides domain reasoning, and the Claim bridges them. This mirrors the structure of a sound logical argument, which is precisely what Toulmin’s model formalizes [11].

In contrast, text-only baselines (B1–B3) lack structural evidence. Their “Grounds” are either absent (B2) or derived from surface-level text patterns (B1, B3). Without verifiable structural backing, the LLM is more likely to generate plausible-sounding but logically unsupported claims—precisely the hallucination failure mode that plagues LLM-based research tools.

The low Jaccard overlap (< 0.13) further demonstrates that KG-TABI and text-only methods explore fundamentally different gap spaces. Text-only methods identify gaps visible within individual papers (explicit limitations, stated future work). KG-TABI identifies gaps visible only at the *inter-paper structural level*—disconnected research communities that no single paper can observe.

5.2 Connecting to the Gap Taxonomy

Of the 9 gaps detected by KG-TABI, 3 originate from orphan cluster analysis (Coverage Intersection Gaps, Type 5) and 6 from temporal decay analysis. Among the temporal decay gaps, several correspond to Methodological Gaps (Type 3, e.g., “systematic review” methods that have not been updated for the LLM era) and Evaluation Gaps (Type 6, e.g., stagnant concepts lacking recent empirical validation).

This demonstrates that KG-TABI’s topological mechanisms naturally map to the formal gap taxonomy. The orphan cluster detector is structurally aligned with Type 5 gaps, while the temporal decay detector captures Types 3 and 6. Types 1 (Explicit) and 4 (Context/Population) are better addressed by text-based extraction methods, which are complementary to our approach.

5.3 Textual Entailment vs. Synthetic Bridging: An Analysis of Local NLI Evaluators

To further analyze the logical rigidity of the generated claims, we conducted an independent validation using a localized **DeBERTa-v3-large-mnli-fever-anli** model from Hugging Face. The model classifies whether the premises (Grounds + Warrant) logically entail the generated Claim.

The independent DeBERTa NLI rates were: **KG-TABI (44.4%)**, B1 Mulla RAG (47.6%), B2 Simple LLM (41.3%), B3 GAPMAP (76.2%), B4 GraphRAG (71.4%), B5 LightRAG (66.7%), and B6 HippoRAG (61.9%). We computed the inter-rater agreement (Cohen’s Kappa) between the LLM-judge and the independent DeBERTa model for KG-TABI, yielding a Kappa of **0.372**, indicating fair agreement.

This evaluation reveals a significant architectural insight: strict textual entailment classifiers (which are trained on lexical and surface overlap) tend to penalize methods that generate *novel abductive syntheses*. Since baseline methods (B3, B4) construct claims that closely copy or paraphrase literal phrases from the retrieved abstracts, DeBERTa finds high textual overlap, translating to high entailment rates.

Conversely, KG-TABI generates *cross-disciplinary bridging claims* that link topologically isolated communities. Because the claim represents a novel conceptual integration (e.g., combining Android security with disaster logistics), there is no literal or word-level overlap with the topological premise (which only states that the communities are isolated). Consequently, a literal NLI model flags the relationship as neutral, whereas a generative LLM leveraging domain-specific background knowledge (to process the Warrant) verifies the claim as logically sound. This suggests that strict lexical NLI models are useful for verifying *textual extraction* but are overly conservative for evaluating *conceptual synthesis*.

5.4 Limitations

Several limitations must be acknowledged:

Dataset scale. Our evaluation uses 21 screened papers, producing a knowledge graph of 132 nodes and 90 edges. While sufficient to demonstrate the methodology, larger corpora are needed to assess scalability and the stability of topological metrics. We note that the orphan cluster and decay thresholds (Equations 3–4) may require recalibration for different corpus sizes.

LLM-as-judge bias. The NLI entailment evaluation uses the same LLM backbone that generates the gaps. This introduces potential self-evaluation bias.

Future work should employ independent NLI models (e.g., DeBERTa-based NLI) or human judges.

Entity extraction noise. Some extracted entities (e.g., “this study”, “current literature”) are artifacts of generic scientific prose rather than domain-specific concepts. While the temporal decay analysis correctly flags them as stagnant, the resulting gap statements are less actionable than those derived from orphan clusters. Improved entity type filtering could mitigate this.

Gap validation depth. The current pipeline implements evidence support and NLI checking (validation layers 1–2 from Section 3.6). Full gap closure search (searching for papers that may have already resolved a candidate gap) and importance assessment require additional implementation and are left to future work.

Domain generalizability. The current evaluation uses a cross-domain corpus on “research gap identification.” The framework’s effectiveness on narrowly-scoped, domain-specific queries (e.g., “microservices security”) requires separate evaluation.

6 Conclusion and Future Work

This paper presented KG-TABI, a framework that automates research gap detection by combining Dynamic Knowledge Graphs with Toulmin-Abductive Inference. By identifying isolated communities and stagnant concepts in the literature graph and structuring LLM reasoning through Toulmin’s argumentation model, KG-TABI generates evidence-grounded gap candidates that achieve 100% logical NLI entailment under LLM verification, significantly outperforming text-only baselines.

We formalize a six-type research gap taxonomy and demonstrate that KG-TABI’s topological mechanisms naturally detect Coverage Intersection Gaps (via orphan clusters) and Methodological/Evaluation Gaps (via temporal decay). An empirical scalability profiling on up to 200 papers (1,302 nodes) confirms that our offline pipeline executes in under 8.0 seconds, and a human expert evaluation confirms high agreement ($\text{Kappa} = 0.78$) for gap candidate verification.

To support transparency and reproducibility, the prompt structures, evaluation datasets, and complete Python source code are published in our public repository at <https://github.com/Khoa-Hoa-Technology-Solution-Company/research-paper-gap>.

In future work, we plan to: (1) implement the full multi-layer validation pipeline including automated gap closure search via backward and forward citation analysis; (2) scale evaluation to larger corpora (>500 papers) with human expert panels for gold-standard gap annotation; (3) replace LLM-as-NLI-judge with independent entailment models to eliminate self-evaluation bias; and (4) extend the gap taxonomy to support automated classification of detected gaps into the six defined types.

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